

Octa Byte AI Pvt Ltd

CIN: U62099KA2025PTC201620

Registered office address: WeWork Prestige Cube, Site no :26, Laskar Housur Rd, Adugodi, Bangalore South, Bangalore -560030, Karnataka

Dear Candidate,

Congratulations on being shortlisted for this round. The objective of this round is to understand hands on work done and your ability to justify the approach. This should take around 9-12 hrs. Some points to make this quick for you:

1. Application logic is not important. You can choose any readymade project from net.
2. Take happy path first
3. You can note good practices to differentiate yourself at specific point.
4. You can also note what extra code would help.

This is comprehensive assignment. We have kept it like that since the role demands it. This needs end to end ownership, skills and willingness to take it up. The experience will be rewarding once you join our startup.

Part 1: Infrastructure Provisioning

1. Using Terraform, provision the following infrastructure on AWS (preferred) or another cloud provider:
 - VPC with public and private subnets
 - EC2 instances or ECS/EKS for application hosting
 - RDS for PostgreSQL database
 - Security groups with appropriate rules
 - Load balancer for the frontend
2. Include variables.tf for configurable parameters
3. Implement proper state management
4. Add outputs for key resources

Part 2: Deployment Automation

1. Create CI/CD pipelines using GitHub Actions/GitLab CI/Jenkins (your choice) that:
 - Run tests on PR creation
 - Build and push Docker images to a registry on merge to main
 - Deploy to staging environment
 - Include a manual approval step for production deployment
2. The pipeline should:
 - Run unit and integration tests
 - Scan for vulnerabilities in dependencies and containers
 - Notify on failures (Slack/email)

Part 3: Monitoring and Logging

1. Set up monitoring that includes:
 - Infrastructure metrics (CPU, memory, disk)
 - Application metrics (request rate, error rate, latency)

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- Database metrics
- 2. Configure centralized logging for:
 - Application logs
 - System logs
 - Access logs
- 3. Create at least two meaningful dashboards

Part 4: Documentation and Best Practices

- 1. Write clear README.md that explains:
 - How to set up and run the infrastructure
 - Architecture decisions
 - Security considerations
 - Cost optimization measures
- 2. Implement at least one:
 - Secret management
 - Backup strategy

Deliverables:-

- 1. Share GitHub repository
- 2. Approach documentation
- 3. Challenges faced during the assignment and resolutions made.

All the best !!

Timeline – 3 days.